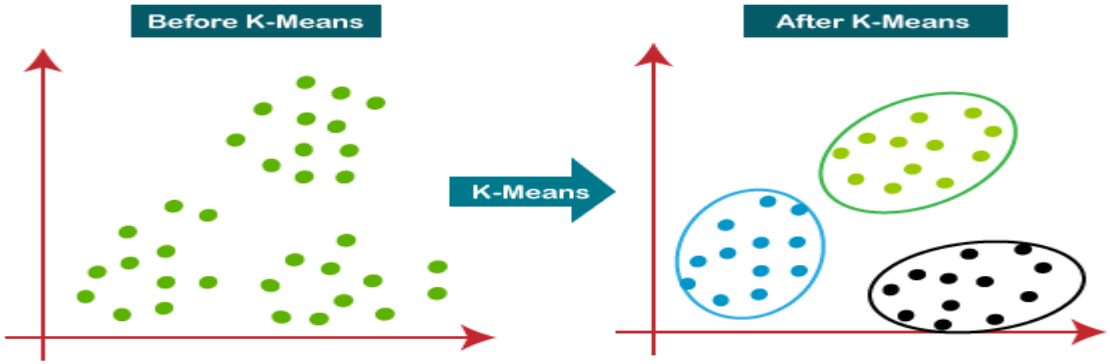
K-Means Clustering

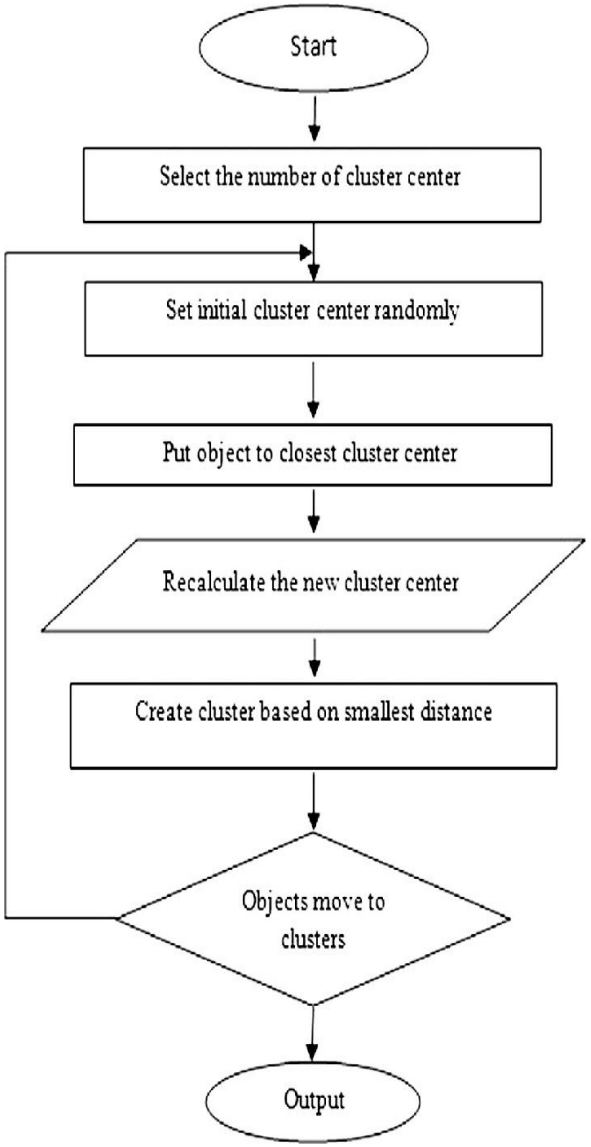
Introduction

- Unsupervised Learning Technique.
- Groups similar data points into K clusters.
- Identifies patterns, relationships, and structure in data.
- Simple, efficient, and scalable.
- Widely used in data mining, image segmentation, customer segmentation, etc.



How K-Means Works & Key Concepts

- Initialize K centroids randomly
- Assign data points to closest centroid
- Update centroids based on assigned data points
- Repeat until convergence



Key Concepts:

* Centroids: Cluster centers
* Distance metric: Euclidean, Manhattan, etc.
* K: Number of clusters.
* Initialization methods: Random, K-Means++, etc.

Advantages, Limitations & Applications

Advantages:

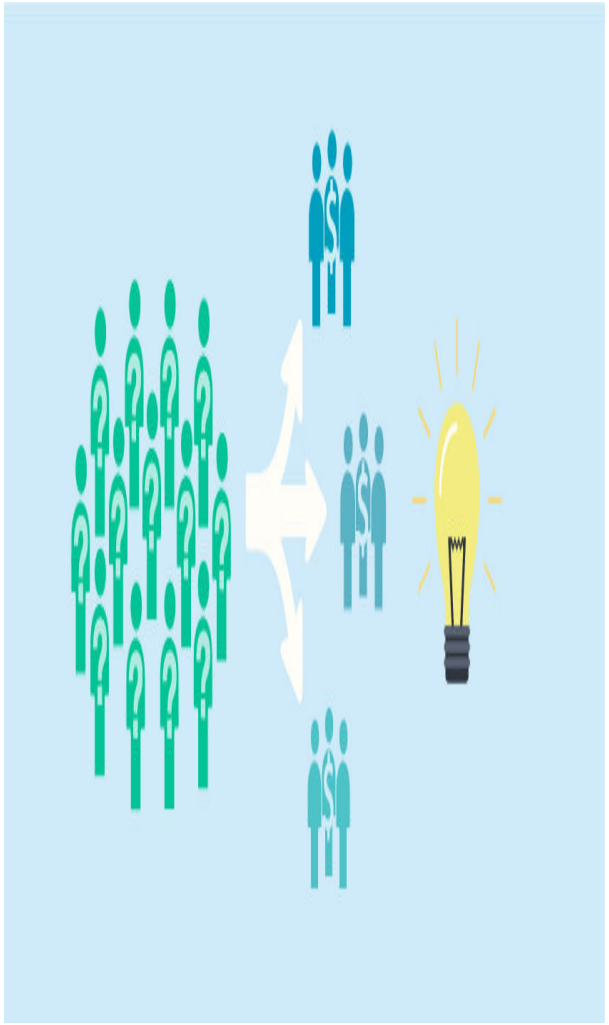
- Simple to implement.
- Efficient computation .
- Handles large datasets.

Limitations:

- Sensitive to initial centroids .
- Assumes spherical clusters.
- Not suitable for complex relationships.

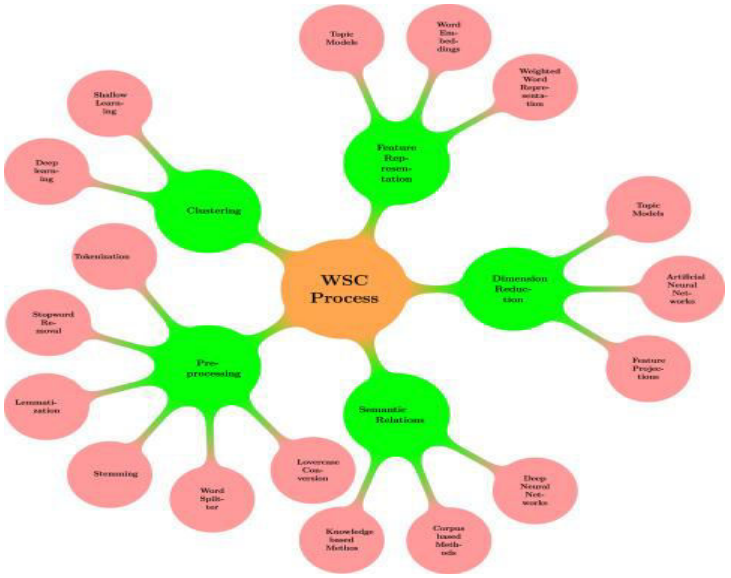
Applications:

- Customer segmentation.
- Image compression.
- Gene expression analysis.
- Recommendation systems.



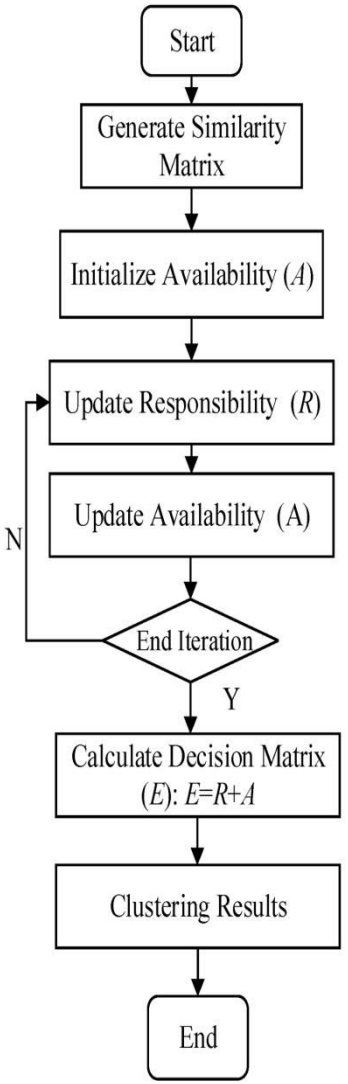
Introduction to Affinity Propagation Clustering

- Identifying Exemplars and Clusters
- Unsupervised clustering algorithm
- Identifies exemplars (representative data points)
- Clusters data points around exemplars
- No predefined number of clusters needed
- Handles noisy and outlier data



How Affinity Propagation Works & Key Concepts

- Sends messages between data points to identify exemplars
- Two types of messages: responsibility and availability
- Iterative process until convergence
- Exemplars become cluster centers



Key Concepts:

- Similarity matrix: measures similarity between data points
- Preference value: controls number of exemplars
- Damping factor: controls convergence rate
- Convergence criteria: iteration limit or change threshold

Advantages, Limitations & Applications

Advantages:

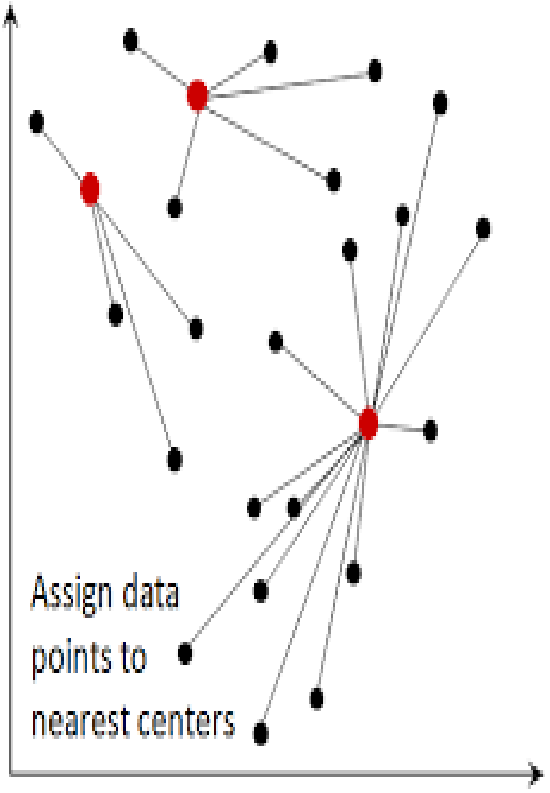
- Identifies representative data points (exemplars)
- Handles noisy and outlier data
- Flexible clustering approach
- No predefined number of clusters needed

Limitations:

- Computationally expensive
- Sensitive to preference value
- Not suitable for very large datasets

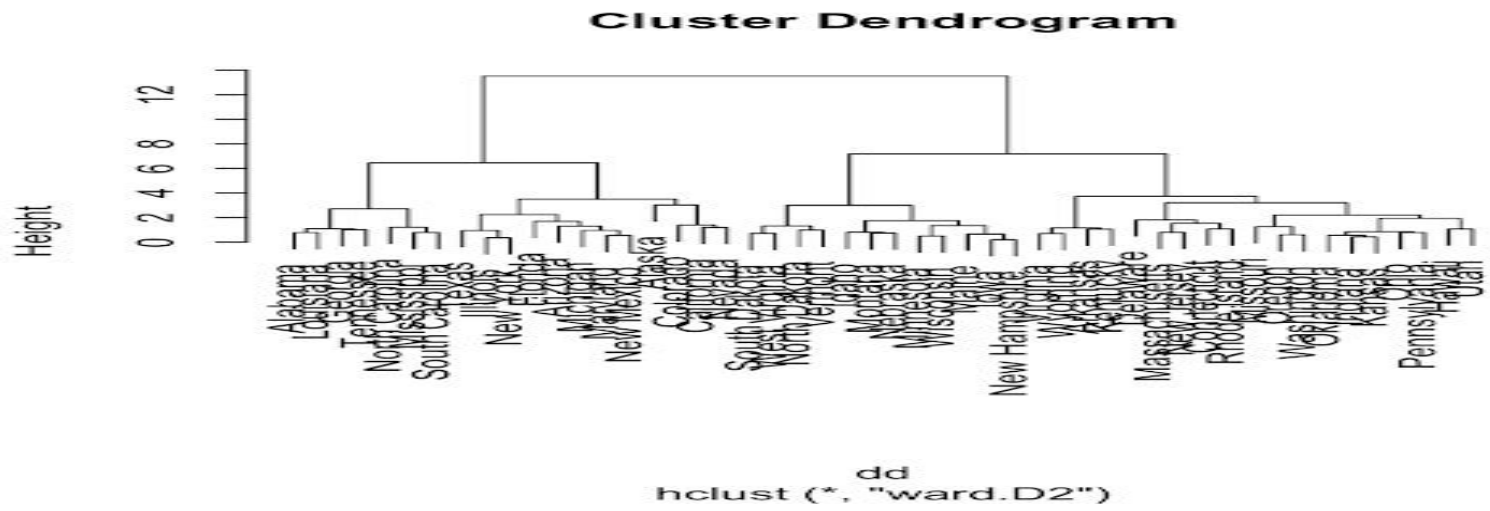
Applications:

- Image segmentation
- Gene expression analysis
- Customer segmentation
- Recommendation systems



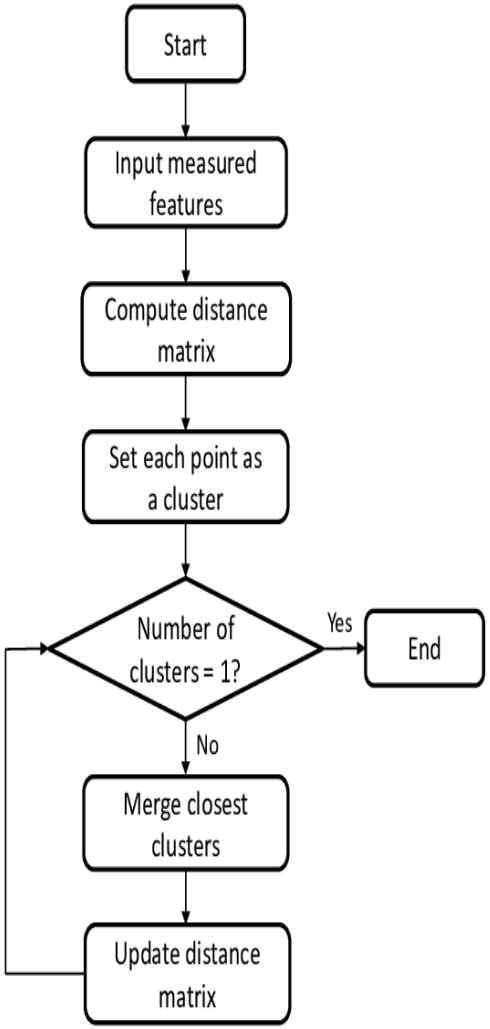
Introduction to Hierarchical Clustering

- Nested Clustering Structure
- Unsupervised clustering algorithm
- Builds hierarchy of clusters
- No predefined number of clusters needed
- Suitable for exploratory data analysis
- Visualizes cluster relationships



How Hierarchical Clustering Works & Key Concepts

- Agglomerative (bottom-up) or Divisive (top-down)
- Distance metric: Euclidean, Manhattan, etc.
- Linkage criterion: Single, Complete, Average, Ward's
- Merge or split clusters based on similarity



Key Concepts:

- Dendrogram: visual representation of hierarchy
- Cluster distance: measures similarity between clusters
- Linkage: method for merging or splitting clusters
- Cutoff point: determines number of clusters

Advantages, Limitations & Applications

Advantages:

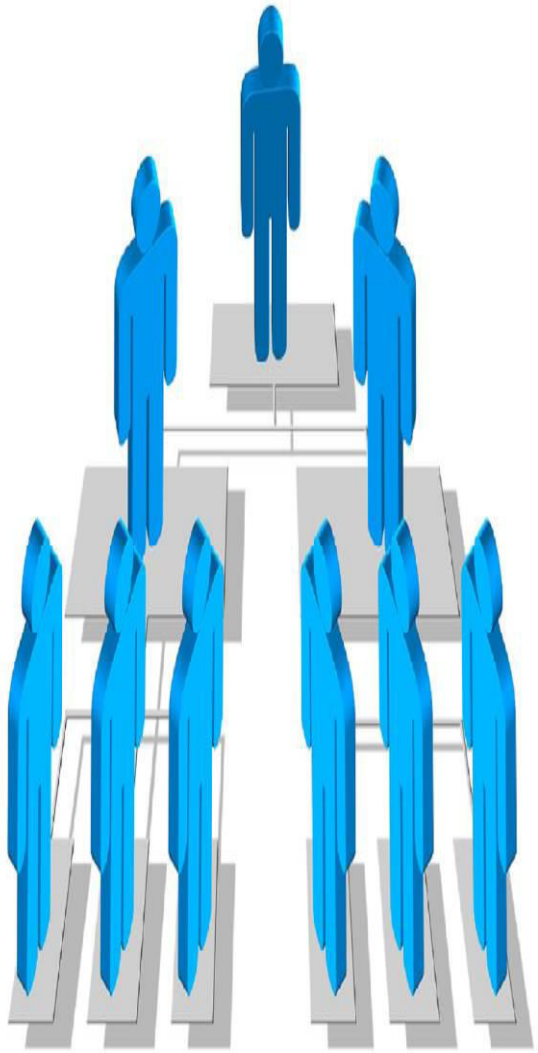
- Provides nested cluster structure
- Visualizes cluster relationships
- No predefined number of clusters needed
- Robust to outliers

Limitations:

- Computationally expensive
- Sensitive to distance metric and linkage
- Difficult to interpret large dendrograms

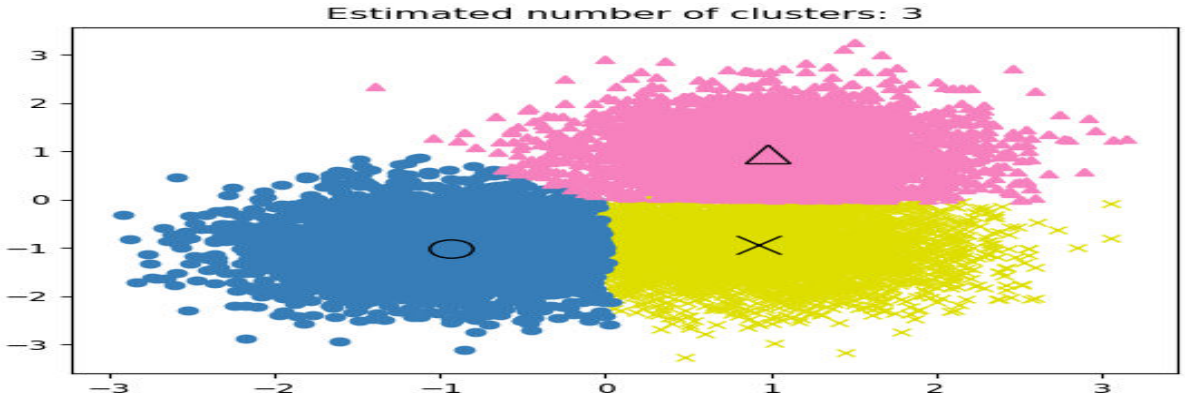
Applications:

- Gene expression analysis
- Image segmentation
- Customer segmentation
- Text classification



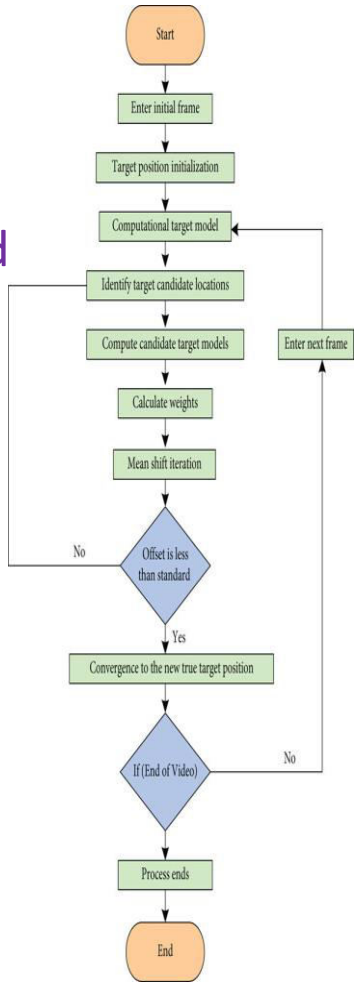
Introduction to Mean Shift Clustering

- Non-Parametric Clustering Technique
- Unsupervised clustering algorithm
- No predefined number of clusters needed
- Identifies clusters of varying shapes and sizes
- Robust to noise and outliers
- Suitable for multimodal distributions



How Mean Shift Clustering Works & Key Concepts

- Iterative process: shift points towards density modes
- Uses kernel density estimation (KDE)
- Bandwidth parameter controls cluster size
- Convergence criteria: iteration limit or convergence threshold



Key Concepts:

- Kernel function: Gaussian, Epanechnikov, etc.
- Bandwidth parameter: controls smoothing
- Density modes: local maxima of density estimate
- Convergence rate: affects computational efficiency

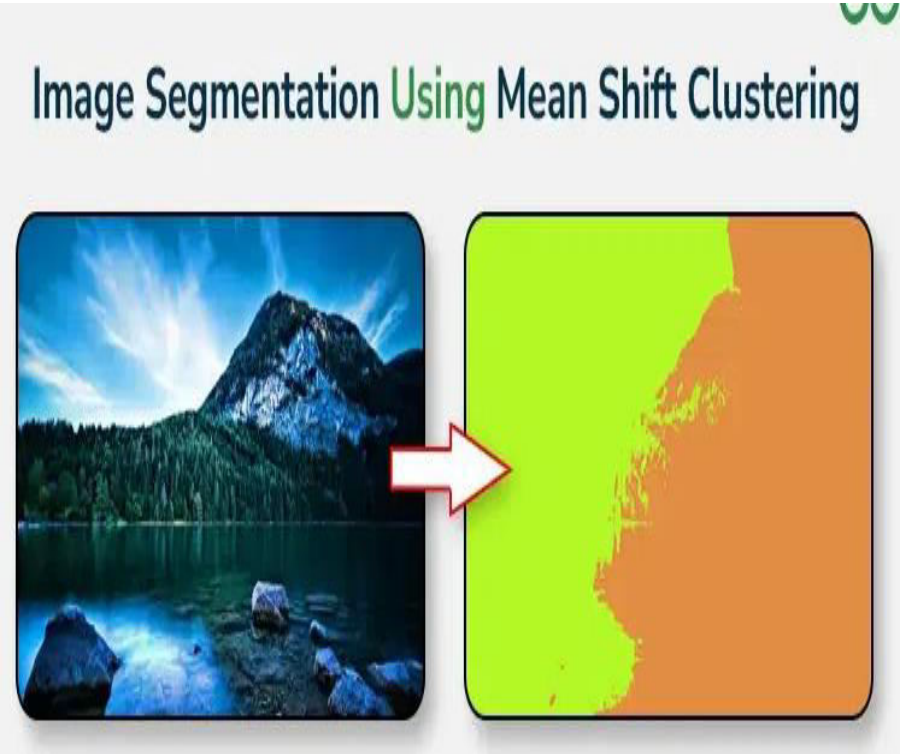
Advantages, Limitations & Applications

Advantages:

- Handles varying cluster shapes and sizes
- Robust to noise and outliers
- No predefined number of clusters needed
- Suitable for multimodal distributions

Limitations:

- Computationally expensive
- Sensitive to bandwidth parameter
- Difficult to interpret clusters

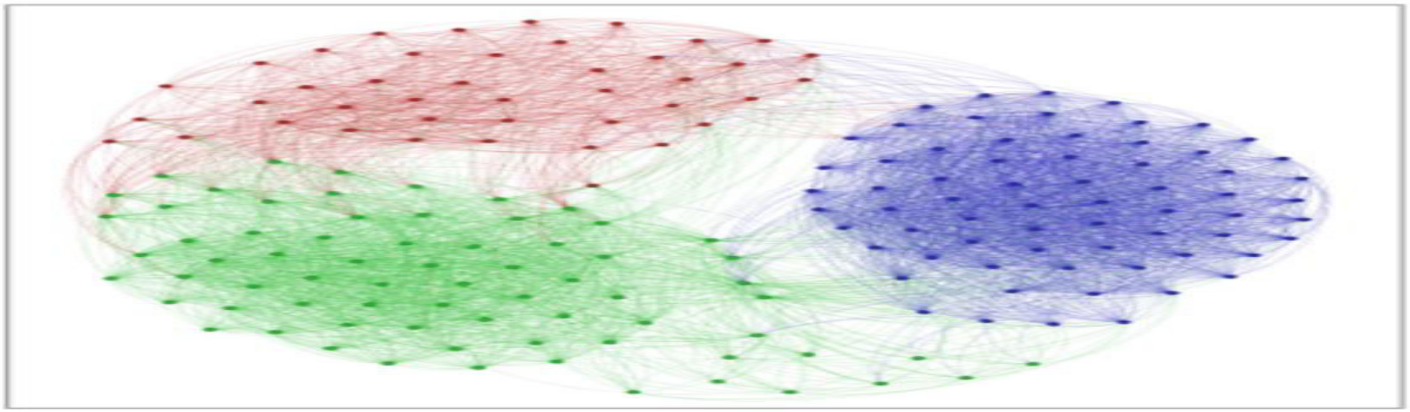


Applications:

- Image segmentation
- Object tracking
- Customer segmentation
- Gene expression analysis

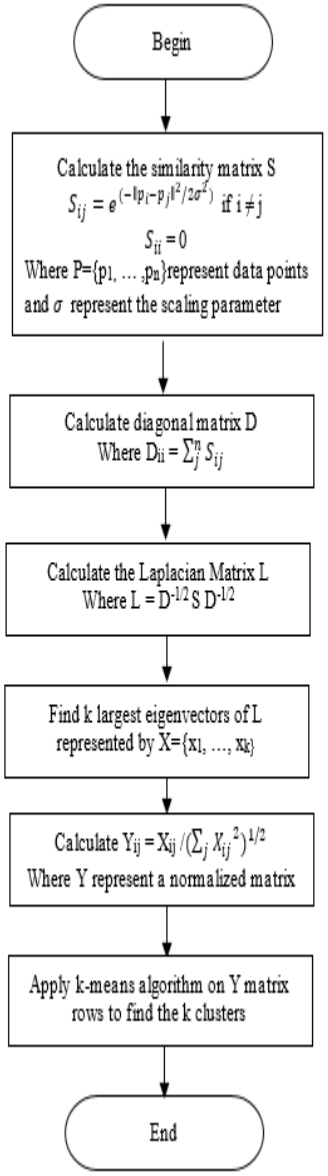
Introduction to Spectral Clustering

- Graph-Based Clustering Technique
- Unsupervised clustering algorithm
- Views data as graph or network
- Identifies clusters as connected components
- Handles non-linear relationships
- Robust to noise and outliers



How Spectral Clustering Works & Key Concepts

- Constructs affinity matrix (similarity graph)
- Computes eigenvectors and eigenvalues
- Applies k-means or other clustering method
- Handles varying cluster sizes and shapes



Key Concepts:

- Affinity matrix: measures similarity between data points
- Laplacian matrix: represents graph structure
- Eigenvectors: represent cluster assignments
- Number of clusters (k): user-defined or estimated

Advantages, Limitations & Applications

Advantages:

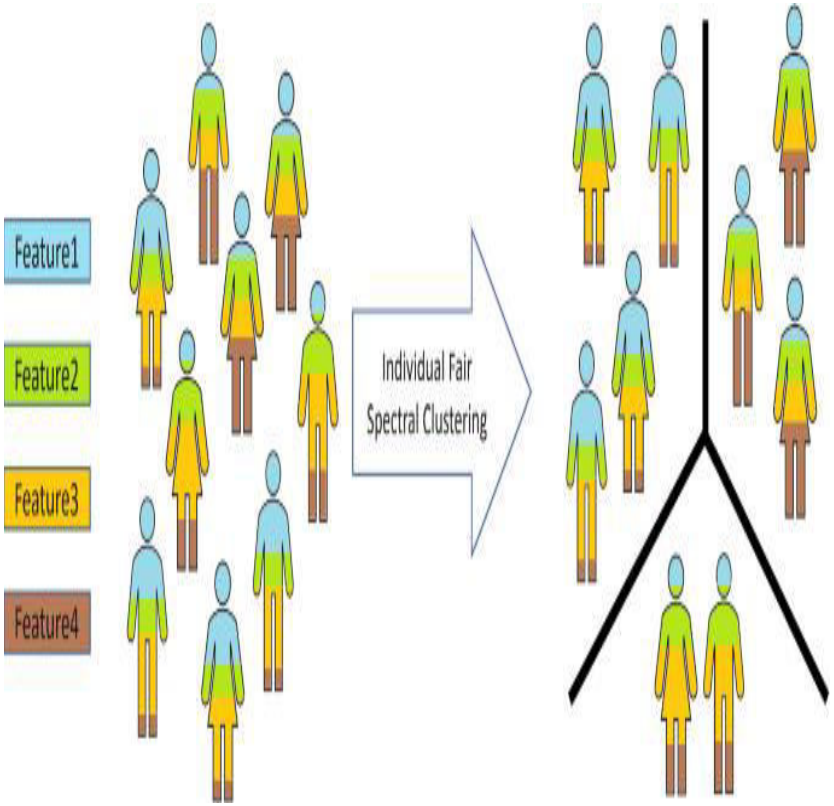
- Handles non-linear relationships
- Robust to noise and outliers
- Identifies clusters of varying sizes and shapes
- Suitable for high-dimensional data

Limitations:

- Computationally expensive
- Sensitive to affinity matrix and k
- Difficult to interpret clusters

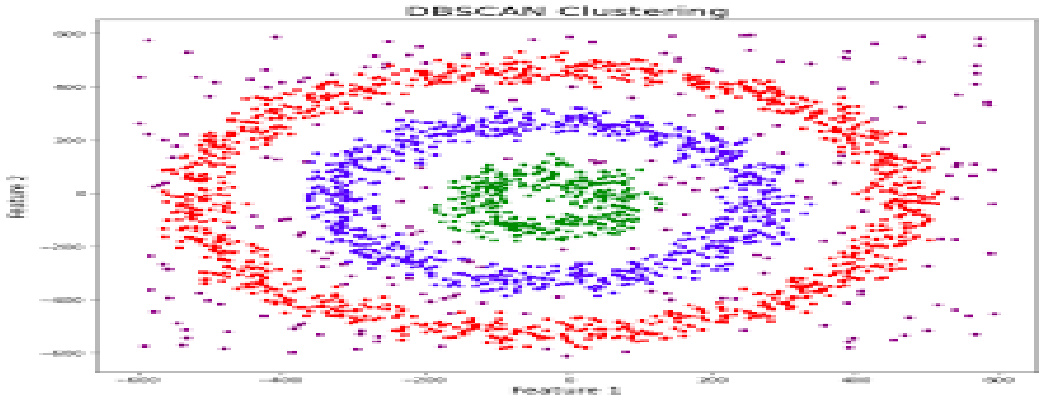
Applications:

- Image segmentation
- Network analysis
- Gene expression analysis
- Recommendation systems



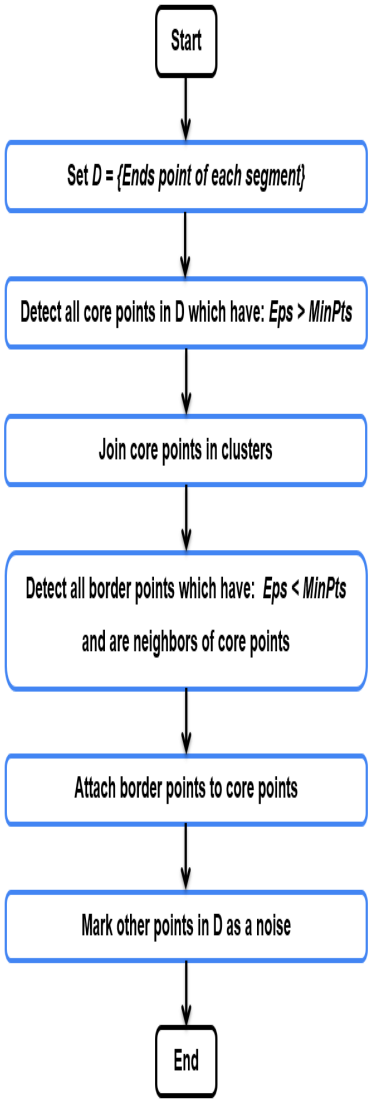
Introduction to DBSCAN Clustering

- Density-Based Clustering Algorithm
- Unsupervised clustering algorithm
- Identifies clusters of varying shapes and sizes
- Handles noise and outliers
- No predefined number of clusters needed
- Robust to density variations



How DBSCAN Clustering Works & Key Concepts

- Finds core points (dense regions)
- Expands clusters from core points
- Handles border points and noise
- Uses ε (epsilon) and MinPts parameters



Key Concepts:

- ε (epsilon): maximum distance between points
- MinPts: minimum points required for dense region
- Core point: point with MinPts neighbors within ε
- Border point: point near core point but not core point
- Noise point: point not part of any cluster

Advantages, Limitations & Applications

Advantages:

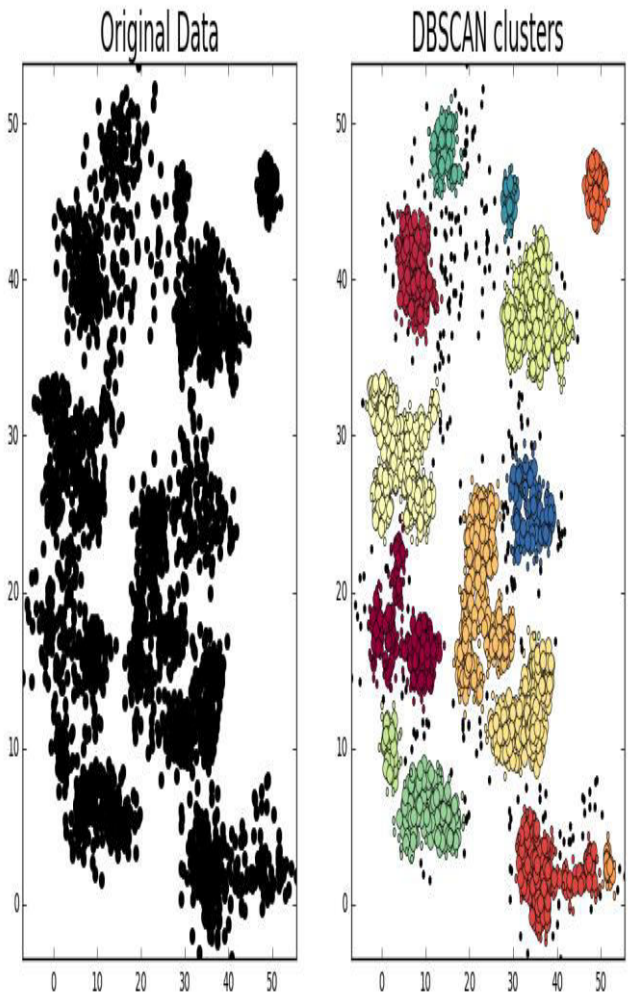
- Handles varying cluster shapes and sizes
- Robust to noise and outliers
- No predefined number of clusters needed
- Handles high-dimensional data

Limitations:

- Sensitive to ε and MinPts parameters
- Computationally expensive
- Difficult to interpret clusters

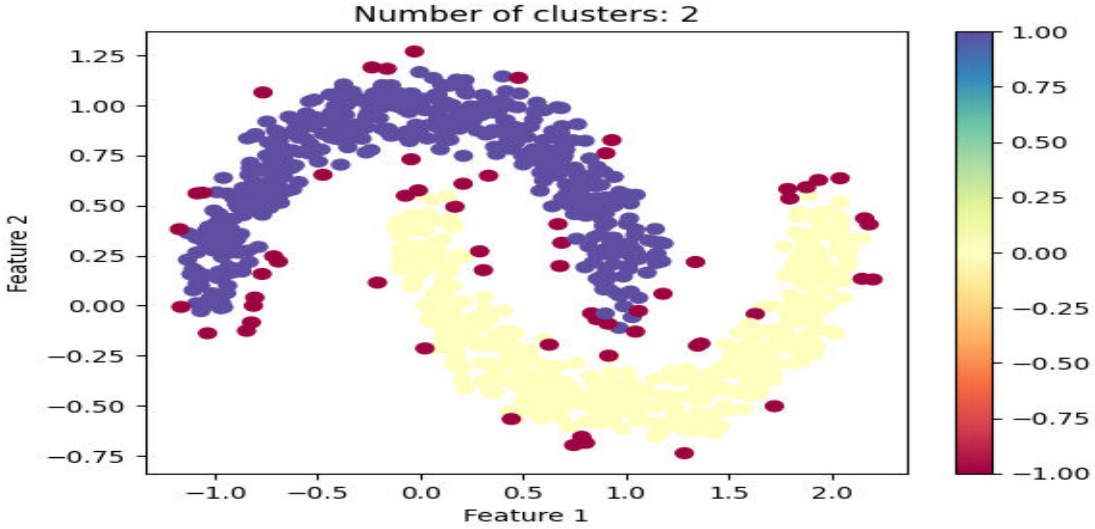
Applications:

- Spatial data analysis
- Image segmentation
- Network analysis
- Anomaly detection



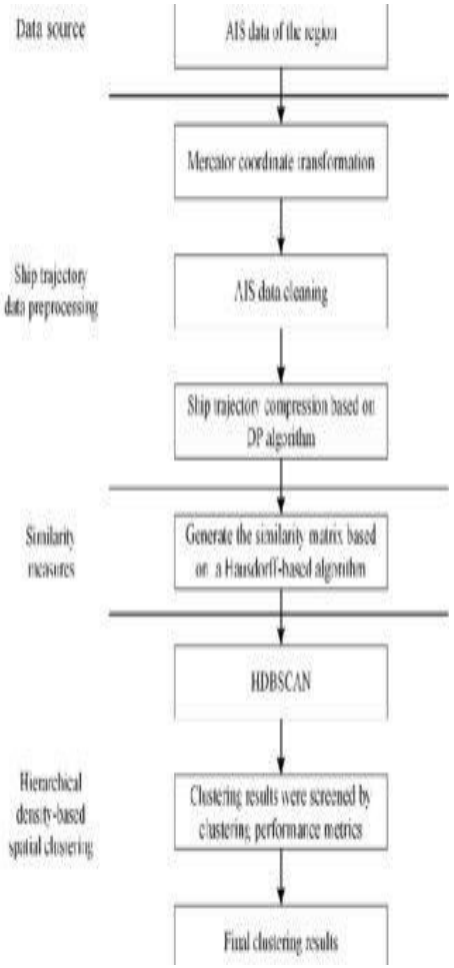
Introduction to HDBSCAN Clustering

- Hierarchical Density-Based Clustering Algorithm
- Unsupervised clustering algorithm
- Identifies clusters of varying shapes and sizes
- Handles noise and outliers
- No predefined number of clusters needed
- Robust to density variations



How HDBSCAN Clustering Works & Key Concepts

- Builds hierarchical cluster tree (dendrogram)
- Uses mutual reachability distance
- Identifies clusters at varying granularity
- Handles varying density clusters



Key Concepts:

- Mutual reachability distance: measures similarity
- Minimum cluster size: controls cluster granularity
- Cluster stability: measures cluster robustness
- Excess of density: identifies cluster boundaries

Advantages, Limitations & Applications

Advantages:

- Handles varying cluster shapes and sizes
- Robust to noise and outliers
- No predefined number of clusters needed
- Handles high-dimensional data

Limitations:

- Computationally expensive
- Sensitive to minimum cluster size
- Difficult to interpret clusters

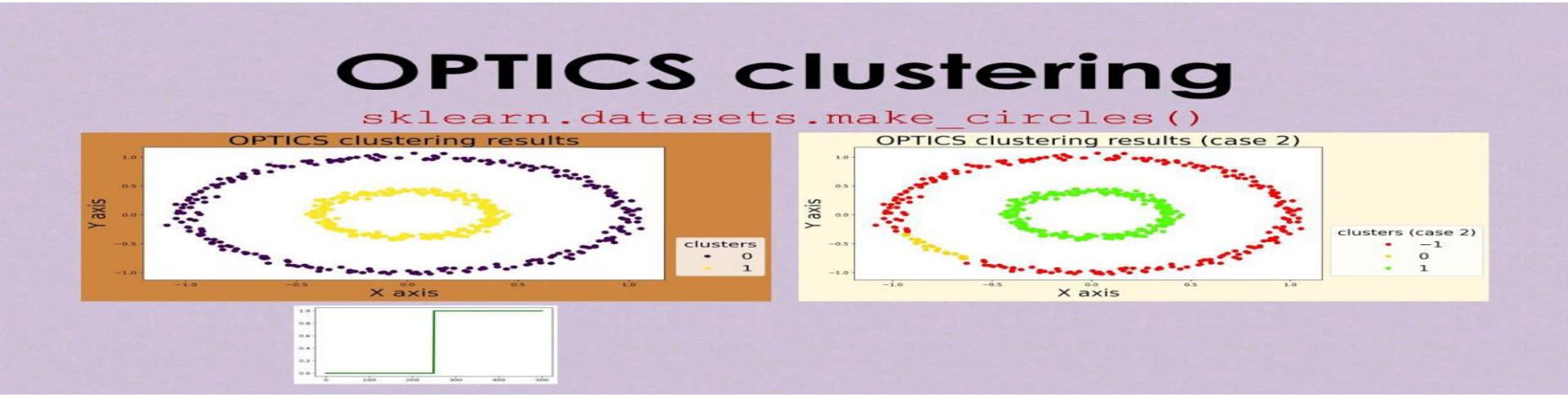
Applications:

- Gene expression analysis
- Image segmentation
- Network analysis
- Anomaly detection



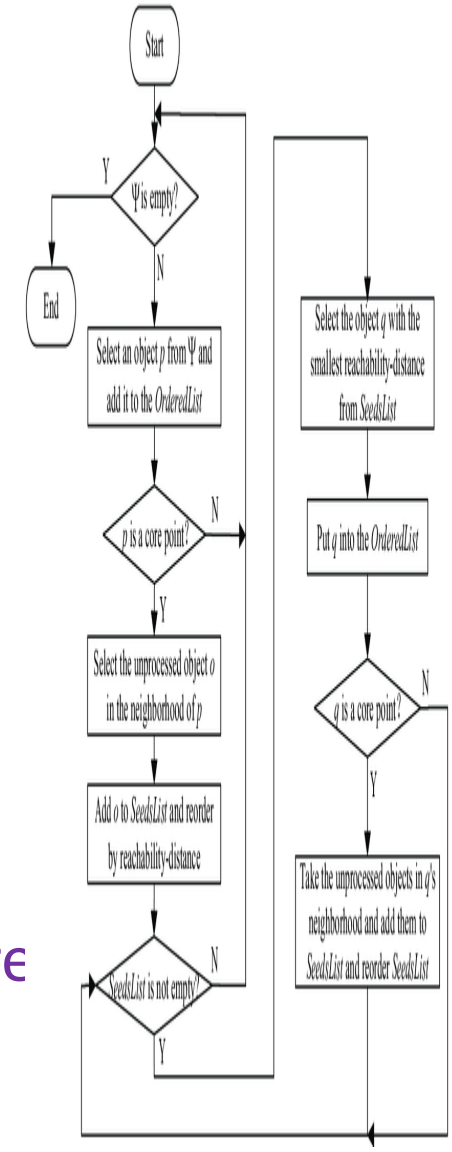
Introduction to OPTICS Clustering

- Ordering Points To Identify Clustering Structure
- Unsupervised clustering algorithm
- Identifies clusters of varying densities
- Handles noise and outliers
- No predefined number of clusters needed
- Robust to density variations



How OPTICS Clustering Works & Key Concepts

- Builds reachability graph
- Computes reachability distance
- Identifies density-based clusters
- Handles varying density clusters



Key Concepts:

- Reachability distance: measures similarity
- Core distance: measures density
- Reachability plot: visualizes cluster structure
- Density-based clustering

Advantages, Limitations & Applications

Advantages:

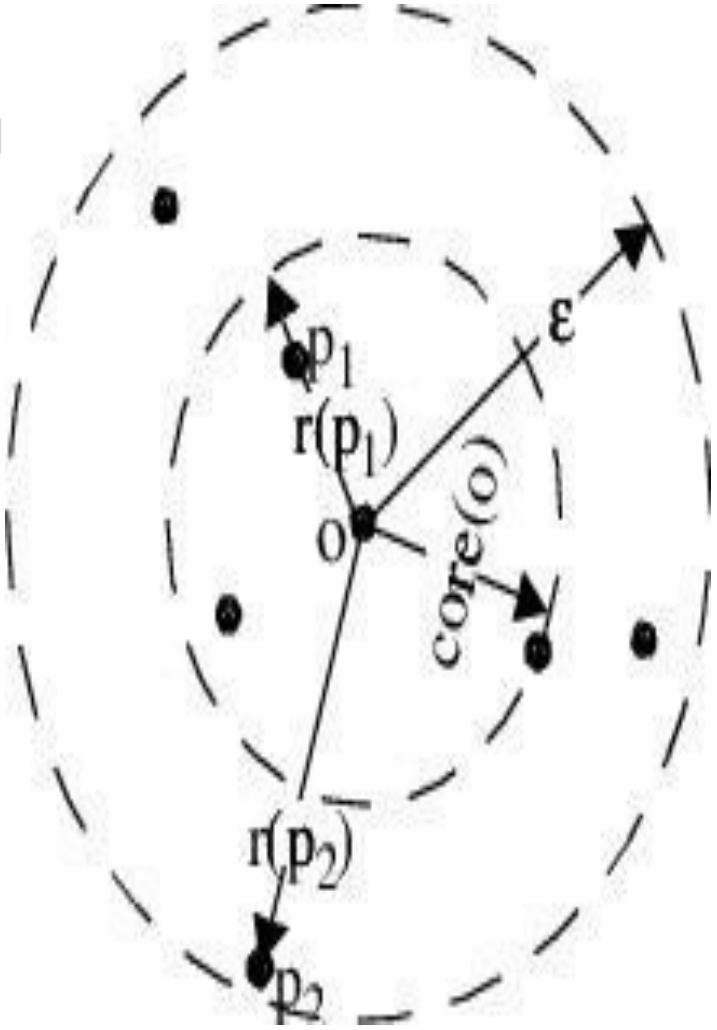
- Handles varying cluster densities
- Robust to noise and outliers
- No predefined number of clusters needed
- Handles high-dimensional data

Limitations:

- Computationally expensive
- Sensitive to parameters
- Difficult to interpret clusters

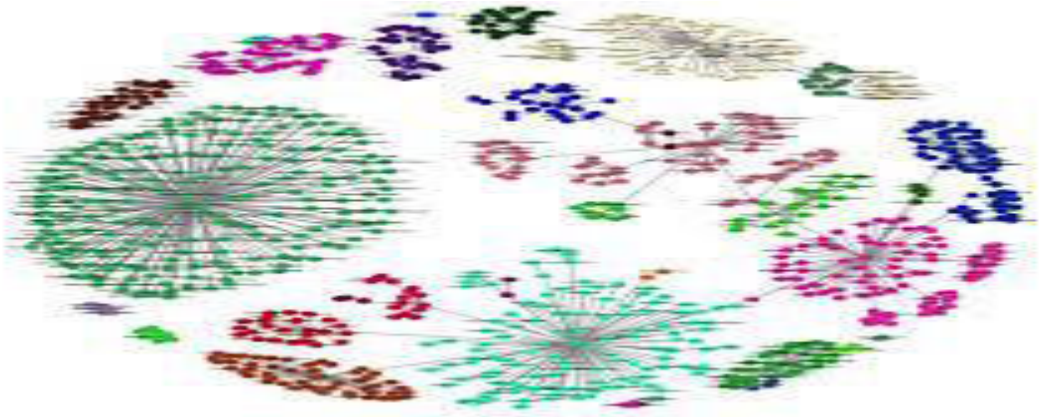
Applications:

- Gene expression analysis
- Image segmentation
- Network analysis
- Anomaly detection



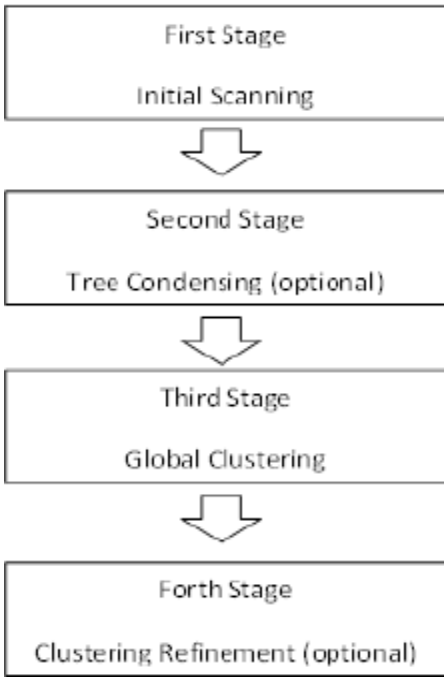
Introduction to BIRCH Clustering

- Balanced Iterative Reducing and Clustering using Hierarchies
- Unsupervised clustering algorithm
- Handles large datasets
- Efficient clustering for incremental data
- Robust to noise and outliers
- Scalable for high-dimensional data



How BIRCH Clustering Works & Key Concepts

- Builds CF (Clustering Feature) tree
- Stores cluster summaries
- Iteratively refines clusters - Handles outliers and noise



Key Concepts:

- CF tree: hierarchical representation
- CF entries: cluster summaries
- Threshold T: controls cluster size
- Branching factor B: controls tree width

Advantages, Limitations & Applications

Advantages:

- Efficient for large datasets
- Handles incremental data
- Robust to noise and outliers
- Scalable for high-dimensional data

Limitations:

- Sensitive to parameters (T, B)
- Difficult to interpret clusters
- Not suitable for varying density clusters

Applications:

- Data mining
- Image segmentation
- Network analysis
- Text clustering

